Exam Prep

An open workspace for exam preparation: notes, summaries, and practice material, kept in Markdown so everything stays diff-friendly and searchable.

Status

Scaffolding only — no study content yet. The repo has this README, AGENTS.md, assets/ for diagrams, and the guardrails that keep it Markdown-only.

Markdown-only rule

Everything tracked here is .md, except .gitignore, .gitattributes, the .github//.githooks//.opencode/ config, and images under assets/.

```
Layer  |  Where  |  Purpose
Ignore allowlist  |  .gitignore  |  Non-Markdown files never show up in git status
Pre-commit hook  |  .githooks/pre-commit  |  Rejects the commit locally
CI check  |  .github/workflows/markdown-only.yml  |  Fails the build on any non-allowlisted file
Ruleset  |  branch main  |  Requires the CI check; direct pushes are blocked
Auto-merge robot  |  .github/workflows/auto-merge.yml  |  Arms auto-merge; GitHub merges only when the check is green
Agent rules  |  .opencode/opencode.jsonc, AGENTS.md  |  Stops the agent creating other file types
```

Enable the hook once per clone:

```
git config core.hooksPath .githooks
```

Branch flow

main is locked. Nothing is pushed to it directly — GitHub rejects the push.
Every change goes through a side branch and a pull request:

```
git switch -c notes/graphs          # one branch per change
git commit -am "graphs: add traversal notes"
git push -u origin notes/graphs
gh pr create --fill                 # robot arms auto-merge on open
```

The auto-merge workflow queues the PR for squash merge. GitHub then merges it itself, but only after the required markdown-only check passes. If the branch contains a file that is not Markdown (or an allowlisted exception), that check fails and the PR stays open — no bad file can reach main.

Merged branches are deleted too: immediately for manual merges, and by an hourly sweep for robot merges, because GitHub suppresses closed events for PRs that auto-merge merges.

To review a PR without merging, open it as a draft; the robot skips drafts.

One exception: a PR that changes .github/workflows/ cannot be merged by

the robot. GitHub forbids GITHUB_TOKEN from merging workflow changes (it can never hold the workflows permission), so merge those yourself:

```
gh pr merge <number> --squash --delete-branch
```

The branch cleanup still runs either way.

Planned structure

```
exam-prep/
├── README.md
├── notes/          # one file per topic, written while studying
├── subjects/       # per-subject index pages linking to notes
├── practice/       # questions, past papers, worked solutions
└── cheatsheets/    # one-page condensed revision sheets
```

Each folder gets an index.md as a table of contents, so the repo stays navigable on GitHub without any tooling.

How to use this repo

1. Capture as you go. Write notes in your own words, one topic per file.
2. Link, don't duplicate. Index pages link to notes; a fact lives in one place.
3. Practice beats rereading. Add attempts and corrections to practice/, including the mistakes — those are the valuable part.
4. Condense late. Cheatsheets are written in the final pass, from the notes, not copied from a textbook.
5. Commit often. Small commits with a subject prefix, e.g. git commit -m "algorithms: add graph traversal notes".

Progress

Subject	Notes	Practice	Cheatsheet	Confidence
add subjects here	–	–	–	–

Conventions

- Markdown only (.md), UTF-8, one topic per file.
- File names: lowercase, hyphenated, e.g. binary-search-trees.md.
- Dates in ISO format (2026-10-25) where they matter.
- Keep line lengths reasonable so diffs stay readable.

Public repository. Shared for reference; corrections and suggestions are welcome.